WEEK 2: RISK QUANTIFICATION AND ANALYSIS

Virtual Construction Project | Quantitative Risk Register, Scoring Matrix, Scenario Analysis and Treatment Thresholds

Prepared for: Week 2 Risk Analysis Task

Purpose: Convert previously identified construction-project risks into comparable numerical scores, prioritise treatment, and demonstrate quantitative uncertainty analysis.

Important note on data: This is a worked academic example. The risk events, probabilities, durations, and financial impacts are illustrative assumptions for a virtual construction project, not measured historical project data. Replace them with Week 1 risks and project estimates if those values are available.

Executive Summary

Ten risks were evaluated using a 5×5 probability–impact matrix. The highest initial scores are material price escalation (16/25), labour shortage/productivity loss (16/25), and design errors/rework (15/25). These should receive early management attention because they can affect the project budget, critical path, and quality. The report also demonstrates expected monetary value (EMV), a triangular-distribution Monte Carlo model, sensitivity analysis, and explicit treatment thresholds.

The quantitative results are decision-support estimates rather than forecasts. They are only as reliable as the assumptions entered; risk owners should validate probability, cost ranges, dependencies, and schedule effects before using them for real project decisions.

1. Project Context and Method

The virtual project is assumed to involve design coordination, procurement, site preparation, construction, inspections, and handover. Risks are assessed across cost, schedule, quality, safety, compliance, and information systems.

Method A — Probability–Impact Score

Probability (P) and impact (I) are each rated from 1 to 5. The base score is **Risk Score = P × I**, giving a range of 1–25. This is a prioritisation tool, not a probability of loss or a monetary value. Probability bands: 1 = very unlikely (≤10%); 2 = unlikely (11–20%); 3 = possible (21–40%); 4 = likely (41–60%); 5 = very likely (>60%). Impact is judged on the most serious credible consequence across cost, schedule, quality, safety, and compliance.

Method B — Expected Monetary Value (EMV)

For a simple single-impact estimate, **EMV = probability of occurrence × monetary impact**. For this example, the monetary range is represented by a triangular distribution (minimum, most likely, maximum) and a risk-event probability determines whether the event occurs in each simulated trial.

Method C — Monte Carlo and sensitivity

A 10,000-trial illustrative Monte Carlo simulation randomly samples whether each risk occurs and, if it occurs, samples its financial impact from a triangular distribution. The aggregate results estimate the distribution of additional risk cost under the stated assumptions. Sensitivity is assessed by comparing expected cost contribution by risk. This model assumes independent risk occurrence and excludes correlations, schedule-cost feedback, and mitigation effects; results may therefore understate combined or cascading risks.

Method references

Project Management Institute (PMI) describes probability–impact matrices, risk exposure, and response thresholds. The U.S. Government Accountability Office (GAO) Cost Estimating and Assessment Guide describes sensitivity analysis and Monte Carlo simulation using probability distributions to estimate cost uncertainty. See the references section for public sources.

2. Scoring Scale and Treatment Thresholds

Rating	Probability	Impact guidance
1 — Very low	≤10%	Negligible cost/schedule effect; easily corrected within routine work.
2 — Low	11–20%	Minor disruption; local corrective action, no major milestone effect.
3 — Moderate	21–40%	Manageable rework or delay; may consume float or contingency.
4 — High	41–60%	Material budget or milestone impact; management intervention likely.
5 — Very high	>60%	Critical-path, major quality/compliance, or severe safety consequence.

Score	Priority	Required treatment / escalation
20–25	Critical	Immediate response plan; project manager approval; weekly review; contingency and named owner.
15–19	High	Treatment plan within 5 working days; review weekly until score falls below 15.
8–14	Moderate	Assign an owner; define trigger and mitigation; review at each project meeting.
1–7	Low	Monitor monthly or at milestones; accept only with documented rationale.

3. Quantitative Risk Register

Initial scores are calculated as probability rating × impact rating. Percentages and monetary ranges are illustrative assumptions. Monetary impacts are expressed in Indian rupees (■); duration impacts are days.

ID	Risk	P	I	Score	Event probability	Impact range (■)
R1	Material price escalation	4	4	16	35%	■180,000–■500,000 Most likely: ■300,000
R2	Severe weather / site disruption	3	4	12	25%	■5–■20 Most likely: ■10
R3	Labour shortage / low productivity	4	4	16	30%	■8–■25 Most likely: ■15
R4	Design errors / rework	3	5	15	20%	■50,000–■250,000 Most likely: ■120,000
R5	Late delivery of critical equipment	3	4	12	30%	■7–■28 Most likely: ■14
R6	Health and safety incident	2	5	10	8%	■2–■15 Most likely: ■5
R7	Permit / regulatory approval delay	2	4	8	20%	■5–■30 Most likely: ■12
R8	Quality failure / failed inspection	3	4	12	18%	■20,000–■150,000 Most likely: ■60,000
R9	Equipment breakdown	3	3	9	20%	■3–■12 Most likely: ■6
R10	Cyber/data loss in project systems	2	3	6	12%	■10,000–■80,000 Most likely: ■30,000

Interpretation: R1 and R3 score 16 (high priority), followed by R4 at 15 (high priority). R2, R5, and R8 score 12, indicating moderate priority under the adopted threshold. Safety (R6) scores 10, but its severity should be escalated independently when a credible outcome could cause serious injury; a simple product score must not override statutory safety controls.

4. Risk Matrix

Impact ↓ / Probability →	1	2	3	4	5
5	5	10	15	20	25
4	4	8	12	16	20
3	3	6	9	12	15
2	2	4	6	8	10
1	1	2	3	4	5

Matrix colour guide: green = low (1–7); amber = moderate (8–14); red = high/critical (15–25). The matrix uses the rating scales defined in Section 2.

5. Detailed Risk Justification and Treatment

R1. Material price escalation — Score 16/25

Supplier quotes may expire before purchase orders are placed, or market prices may rise during procurement. A 35% event probability and ■1.8–5.0 lakh impact range are assumed. The risk is high because material costs can affect several work packages simultaneously. Trigger: a critical material quote increases by more than 5% or expires before approval.

Recommended treatment: Lock prices for critical materials; obtain 2–3 supplier quotes; hold a documented contingency.

R2. Severe weather / site disruption — Score 12/25

Heavy rain or extreme heat may stop outdoor activities, damage stored materials, or reduce productivity. The illustrative event probability is 25%, with a 5–20 day impact range. Trigger: forecast conditions exceed the site plan's weather limits or weather delays consume half of schedule float.

Recommended treatment: Use weather allowances, protect materials, resequence indoor work, and maintain float.

R3. Labour shortage / low productivity — Score 16/25

Insufficient skilled labour, absenteeism, or lower-than-planned productivity can delay sequential activities and increase overtime costs. A 30% event probability and 8–25 day range are assumed. Trigger: actual weekly productivity falls below 90% of plan for two consecutive weeks.

Recommended treatment: Confirm crew availability, track weekly productivity, cross-train workers, and use approved backup subcontractors.

R4. Design errors / rework — Score 15/25

Incomplete drawings, coordination clashes, or late design changes can force demolition and rework. The 20% event probability and ■50,000–■2.5 lakh range are illustrative. This is high impact because rework can affect quality and the critical path. Trigger: unresolved high-severity design comments remain at a construction release milestone.

Recommended treatment: Run interdisciplinary design reviews, clash detection, and formal change control before construction.

R5. Late delivery of critical equipment — Score 12/25

A late delivery of a critical component may prevent installation and block follow-on trades. The assumed event probability is 30%, with 7–28 days of potential delay. Trigger: supplier misses a confirmed milestone or fails to provide dispatch evidence.

Recommended treatment: Identify long-lead items early, confirm supplier milestones, and define alternatives for critical components.

R6. Health and safety incident — Score 10/25

Unsafe work at height, electrical work, lifting, or poor housekeeping can cause injury, stoppage, investigation, and legal consequences. The 8% probability is illustrative and should never be interpreted as permission to accept unsafe work. Trigger: any serious near miss, failed safety inspection, or breach of permit-to-work conditions.

Recommended treatment: Conduct toolbox talks, permit-to-work checks, PPE inspections, and stop-work escalation.

R7. Permit / regulatory approval delay — Score 8/25

Incomplete submissions, unclear ownership, or slow authority review may delay site work or occupancy. The 20% event probability and 5–30 day range are assumed. Trigger: a required approval has not been received by its internal due date.

Recommended treatment: Create an approvals register, assign an owner, submit complete applications early, and track authority responses.

R8. Quality failure / failed inspection — Score 12/25

Non-conforming concrete, installation defects, or failed tests may require corrective work and repeat inspection. The assumed probability is 18% and impact range ■20,000–■1.5 lakh. Trigger: a test result falls outside the acceptance criteria or a repeated defect is found.

Recommended treatment: Use inspection and test plans, approved samples, hold points, and corrective-action tracking.

R9. Equipment breakdown — Score 9/25

Plant or equipment failure can stop a dependent activity and incur repair or rental expense. The illustrative event probability is 20%, with 3–12 days of delay. Trigger: missed preventive maintenance, recurring fault, or critical spare unavailability.

Recommended treatment: Schedule preventive maintenance, keep critical spares, and confirm rental backup availability.

R10. Cyber/data loss in project systems — Score 6/25

Loss of drawings, compromised credentials, or ransomware can disrupt coordination and expose project information. The illustrative probability is 12%, with ■10,000–■80,000 impact. Trigger: backup failure, suspicious login, or an access-control alert.

Recommended treatment: Use role-based access, multi-factor authentication, daily backups, and a tested recovery procedure.

6. Expected Monetary Value and Statistical Analysis

EMV is calculated as event probability multiplied by the mean of the triangular impact distribution, where mean = (minimum + most likely + maximum) / 3. This is a simplified expected-cost contribution; it does not capture dependencies or indirect schedule costs.

ID	Risk	Probability	Triangular mean impact	EMV
R1	Material price escalation	35%	■326,667	■114,333
R2	Severe weather / site disruption	25%	■12	■3
R3	Labour shortage / low productivity	30%	■16	■5
R4	Design errors / rework	20%	■140,000	■28,000
R5	Late delivery of critical equipment	30%	■16	■5
R6	Health and safety incident	8%	■7	■1
R7	Permit / regulatory approval delay	20%	■16	■3
R8	Quality failure / failed inspection	18%	■76,667	■13,800
R9	Equipment breakdown	20%	■7	■1
R10	Cyber/data loss in project systems	12%	■40,000	■4,800

Total simplified EMV: ■160,951 across the ten risks. This is the sum of expected individual risk-cost contributions under the stated assumptions, not a guaranteed contingency requirement.

Illustrative Monte Carlo results (10,000 trials)

Mean simulated additional risk cost: ■**160,681**. Median: ■**86,587**. 80th percentile: ■**351,171**. 90th percentile: ■**423,893**. 95th percentile: ■**480,379**. Simulated probability that additional risk cost exceeds ■5,00,000: **3.7%**.

Percentiles are scenario outputs generated from the illustrative inputs. They should not be represented as actual project forecasts. A project contingency may be selected at a chosen confidence level only after validating the distributions, risk dependencies, base estimate, and risk appetite.

7. Sensitivity Analysis

A useful first-pass sensitivity measure is each risk's EMV contribution. The largest contributors deserve further investigation because changing their probability or impact assumptions has a comparatively large effect on expected cost. A proper sensitivity study should then vary one input at a time, document the resulting change in total cost, and use defensible low/most-likely/high estimates rather than arbitrary percentages.

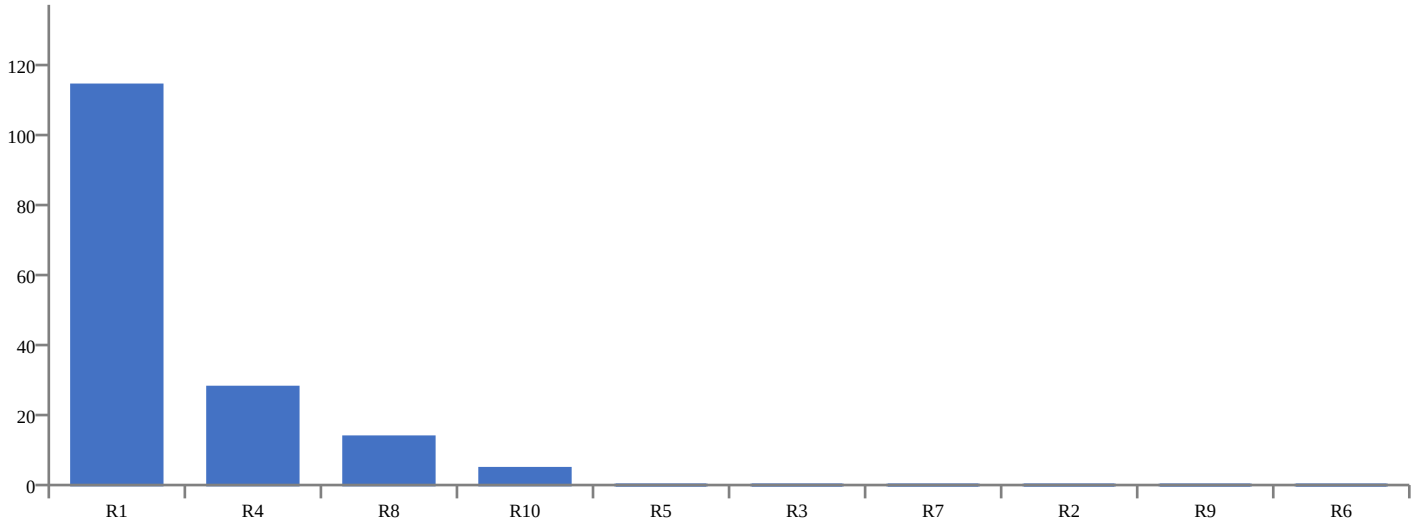

Figure 1. Illustrative expected monetary value by risk (■ thousands).

8. Risk Treatment Plan and Monitoring

Priority	Risk IDs	First actions	Owner / cadence
High (score 15–19)	R1, R3, R4	Confirm quotes and labour plan; conduct design/clash review; create contingency actions and triggers.	Project manager + procurement/design leads; weekly
Moderate (score 8–14)	R2, R5, R6, R7, R8, R9	Track weather and float; expedite long-lead items; enforce safety controls; monitor approvals and inspections; maintain equipment.	Package owners; each coordination meeting; safety checks continuously
Low (score 1–7)	R10	Maintain backups, access controls, and incident response; escalate immediately if an alert occurs.	IT/project controls; monthly and on alerts

Monitoring rules

• Review high risks weekly and all other risks at least monthly or at milestone gates.
• Re-score a risk when probability, impact, trigger status, or treatment effectiveness changes.
• Record each risk owner, due date, evidence, action status, and residual score after mitigation.
• Escalate immediately for serious safety hazards, regulatory breaches, or any event likely to threaten a contractual milestone.
• Do not reduce a score merely because an action has been planned; reduce it only after the action is implemented and evidence supports a lower residual rating.

9. Limitations and Recommendations

This assessment is a structured example, not a substitute for project-specific evidence. Probability estimates should be calibrated using site records, supplier histories, weather data, past productivity, inspection logs, and expert judgement. Risks are not necessarily independent: for example, weather disruption may worsen labour productivity and delay deliveries. The simulation currently assumes independence and models financial impacts only; it does not directly simulate schedule networks, correlated events, safety severity, or the cost of mitigation.

Recommended next step: replace the illustrative risk list with the eight or more risks documented in Week 1, confirm each risk's probability and impact with the project team, and rerun the scoring and simulation. Add a schedule Monte Carlo model if activity durations and dependencies are available.

10. Conclusion

The risk matrix provides a consistent way to prioritise threats, while EMV and Monte Carlo analysis show how uncertain cost impacts can combine into a range of possible outcomes. Material price escalation, labour shortage, and design rework are the leading priorities in this illustrative example. Clear treatment thresholds, assigned owners, measurable triggers, and periodic re-scoring make the assessment actionable rather than a static table.

11. Public References

1. Project Management Institute (PMI), “Risk analysis and management.” https://www.pmi.org/learning/library/risk-analysis-project-management-7070

2. U.S. Government Accountability Office (GAO), “Cost Estimating and Assessment Guide: Best Practices for Developing and Managing Program Costs,” GAO-20-195G (2020). https://www.gao.gov/products/gao-20-195g

3. Project Management Institute (PMI), “Risk management: Monte Carlo simulation cost estimating.” https://www.pmi.org/learning/library/monte-carlo-simulation-cost-estimating-6195

4. Project Management Institute (PMI), “Managing overall project risk.” https://www.pmi.org/learning/library/overall-project-risk-assessment-models-1386

End of report